



Hospital Networking System

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Section BSSE-F-24-A

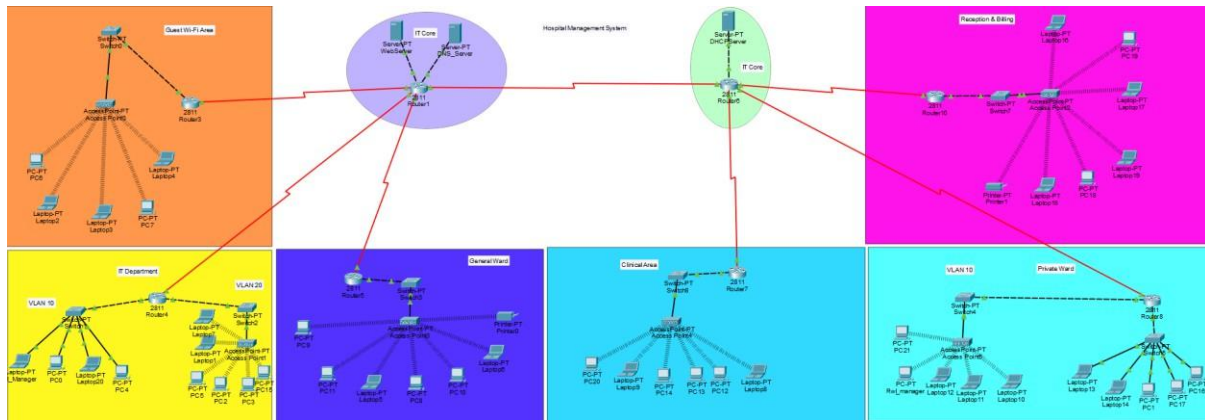
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Contents

1. Project Model Interface	3
2. Executive Summary:	3
3. Project Scope & Requirements	3
4. Network Topology & Design Philosophy:	4
5. IP Addressing Scheme (VLSM Plan)	4
6. Routing Protocol Implementation: OSPF	6
6.1. Why OSPF?	6
7. LAN Design: VLANs & Inter-VLAN Routing	7
8. Server Configurations & Services.....	7
8.1 DHCP Server	7
8.2 DNS & Web Server	8
9. Network Security Implementation.....	8
9.1 Standard & Extended ACLs.....	8
10. Wireless Network Design.....	9
10.1 AP Placement	9
10.2 Security Protocol	9
11. Performance & Quality of Service (QoS)	10
11.1 Latency Testing.....	10
11.2 Packet Loss Analysis.....	10
12. VPN Setting.....	11
Conclusion.....	12

1. Project Model Interface:



2. Executive Summary:

The objective of this project is to design a secure, high-performance, and scalable network for a multi-branch Hospital Management System (HMS). This network integrates various medical departments, secure wireless access for patients/staff, and centralized server-side management. By utilizing advanced networking protocols such as OSPF for internal routing and IPsec VPN for branch-to-head-office connectivity, the system ensures seamless and secure data transmission across the healthcare facility.

3. Project Scope & Requirements :

This report follows the academic and technical requirements for a comprehensive network project:

- **Routing Protocol Implementation:** Deployment of OSPF/EIGRP/BGP across 8 routers.
- **LAN Design with VLANs:** Segmentation of hospital departments using 3+ VLANs and Inter-VLAN routing.
- **WAN Technology:** Implementation of Site-to-Site (S2S) VPN for branch connectivity.
- **Network Security:** Use of Access Control Lists (ACLs), Firewall logs, and Port Security.
- **IP Addressing:** Systematic allocation using Variable Length Subnet Masking (VLSM).

- **Server Configuration:** Setting up DNS, DHCP, and Web servers for hospital services.

4. Network Topology & Design Philosophy:

4.1 Topology Description

The network follows a Star-Bus hybrid topology.

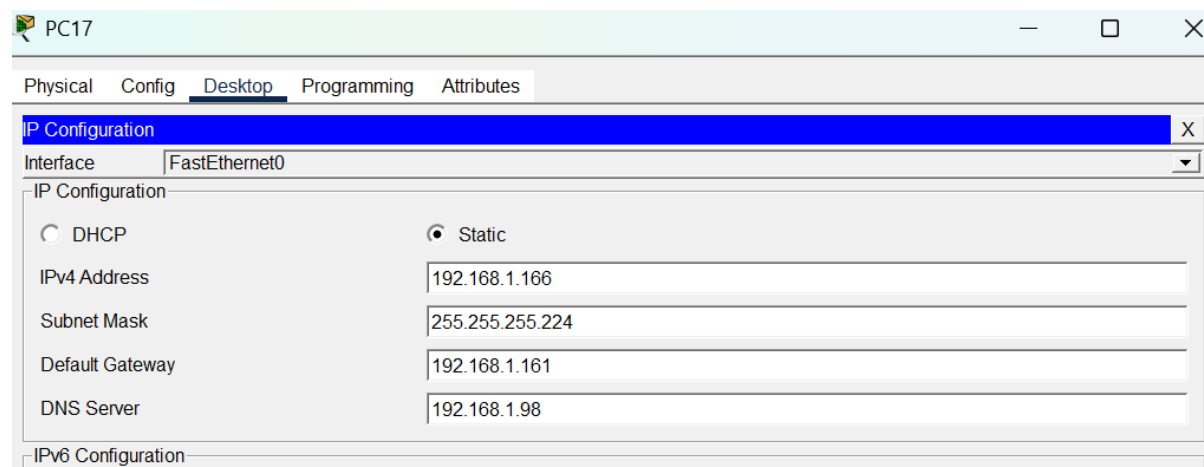
- **Core Layer:** High-speed routers (Router 1 & Router 6) managing the backbone.
- **Access Layer:** Switches connecting end-devices like PCs, Laptops, and Medical Printers.

4.2 Departmental Breakdown

- **IT Department:** Focuses on server management and network monitoring.
- **Clinical/Private Wards:** High-security zones for patient data.
- **Guest Wi-Fi:** Isolated area for public internet access.

5. IP Addressing Scheme (VLSM Plan)

PC:

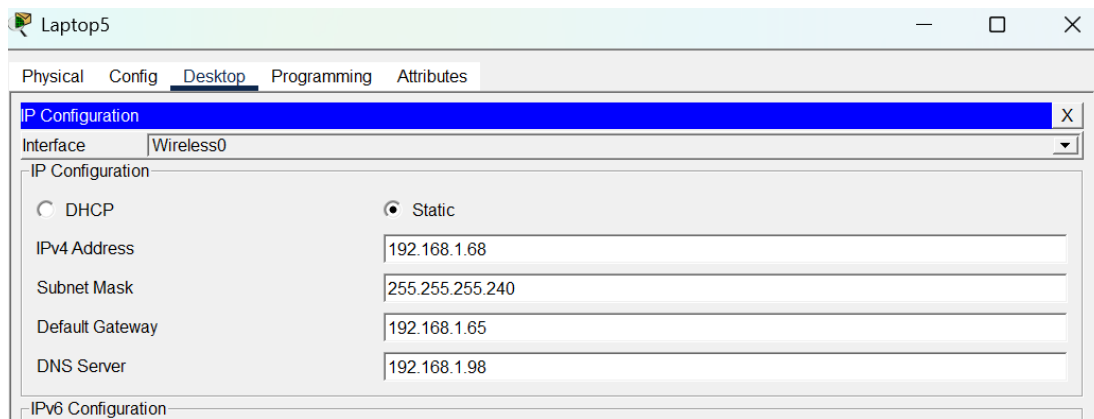


The screenshot shows the 'IP Configuration' window for PC17. The 'Interface' is set to 'FastEthernet0'. Under 'IP Configuration', the 'Static' radio button is selected. The fields are filled with the following values:

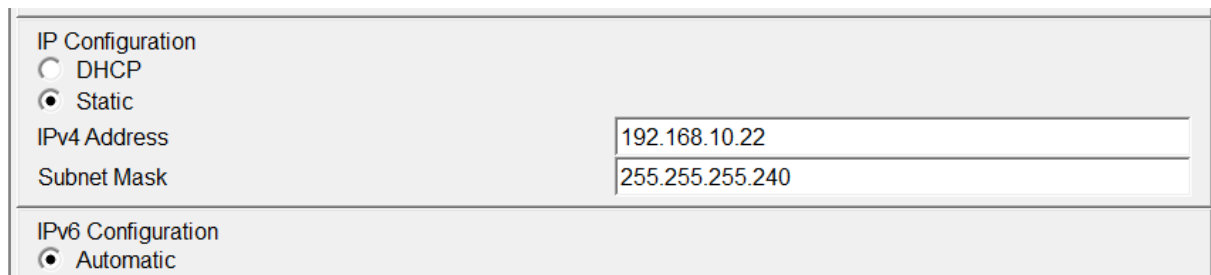
Field	Value
IPv4 Address	192.168.1.166
Subnet Mask	255.255.255.224
Default Gateway	192.168.1.161
DNS Server	192.168.1.98

The 'IPv6 Configuration' section is visible at the bottom but is currently collapsed.

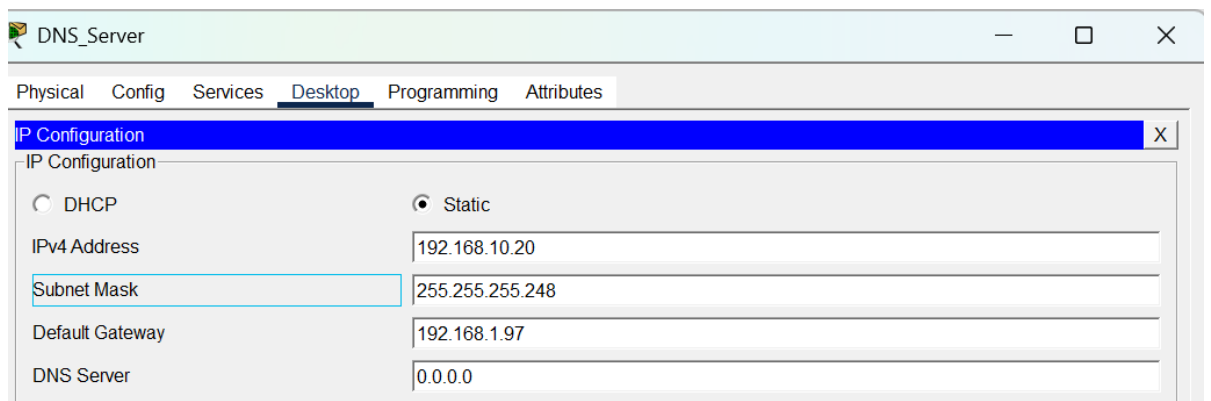
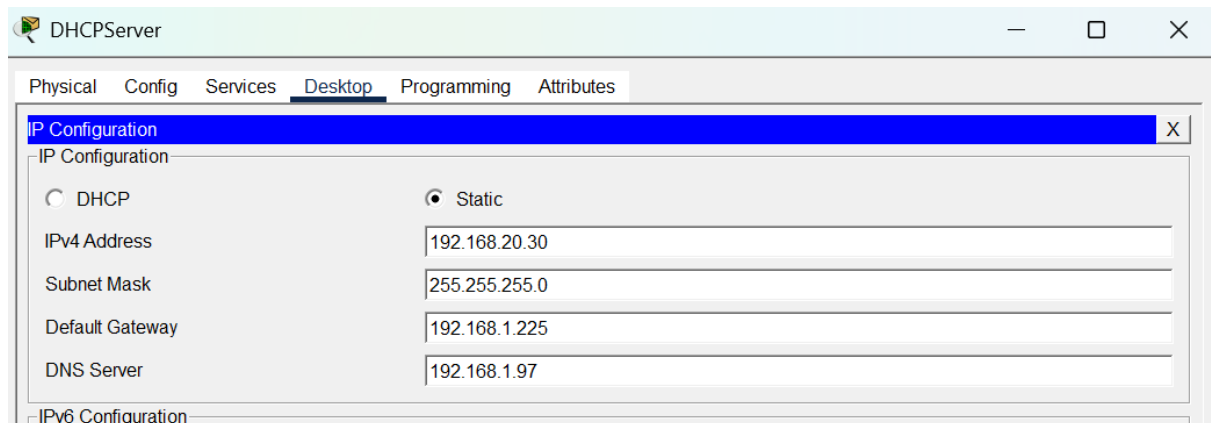
Laptop:



Printer:



Server:



WebServer

Physical Config Services **Desktop** Programming Attributes

IP Configuration X

IP Configuration

☐ DHCP ☒ Static

IPv4 Address: 192.168.10.10

Subnet Mask: 255.255.255.248

Default Gateway: 192.168.1.105

DNS Server: 0.0.0.0

6. Routing Protocol Implementation: OSPF

6.1. Why OSPF?

We chose OSPF (Open Shortest Path First) because of its fast convergence and support for VLSM.

IT Core:

```
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router ospf 1
Router(config-router)#network 192.168.1.224 255.255.255.248 area 0
Router(config-router)#network 192.168.1.232 255.255.255.252 area 0
Router(config-router)#network 192.168.1.236 255.255.255.252 area 0
Router(config-router)#network 192.168.1.240 255.255.255.252 area 0
Router(config-router)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

IT Department:

```
Router>enable
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router ospf 1
Router(config-router)#network 192.168.1.236 255.255.255.252 area 0
Router(config-router)#network 192.168.1.128
00:39:02: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.1.241 on Serial0/0/0 from
LOADING to FULL, Loading Done

% Incomplete command.
Router(config-router)#network 192.168.1.128 255.255.255.224 area 3
Router(config-router)#network 192.168.1.160 255.255.255.224 area 3
Router(config-router)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#
```

Copy

Paste

Verification:

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC7	Router3	ICMP		0.000	N	0	(edit)	(delete)
	Successful	WebS...	PC7	ICMP		0.000	N	1	(edit)	(delete)
	Successful	Lapto...	DHCPServer	ICMP		0.000	N	2	(edit)	(delete)

7. LAN Design: VLANs & Inter-VLAN Routing

7.1 VLAN Assignment

- **VLAN 10:** Admin/Management
- **VLAN 20:** Clinical Staff
- **VLAN 30:** Guest Users

7.2 Trunking and Sub-interfaces

We implemented "Router-on-a-stick" for communication between departments.

```
Switch>show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/1, Fa1/1, Fa2/1, Fa3/1 Fa4/1, Fa5/1
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```
Switch>
```

8. Server Configurations & Services

8.1 DHCP Server

Automatically assigns IPs to Ward laptops.

DHCPServer

Physical
Config
Services
Desktop
Programming
Attributes

SERVICES

HTTP
DHCP
DHCPv6
TFTP
DNS
SYSLOG
AAA
NTP
EMAIL
FTP
IoT
VM Management
Radius EAP
PRP

FastEthernet0

serverPool

192.168.1.225

192.168.1.97

192

168

1

0

255

255

255

0

8

0.0.0.0

0.0.0.0

Add

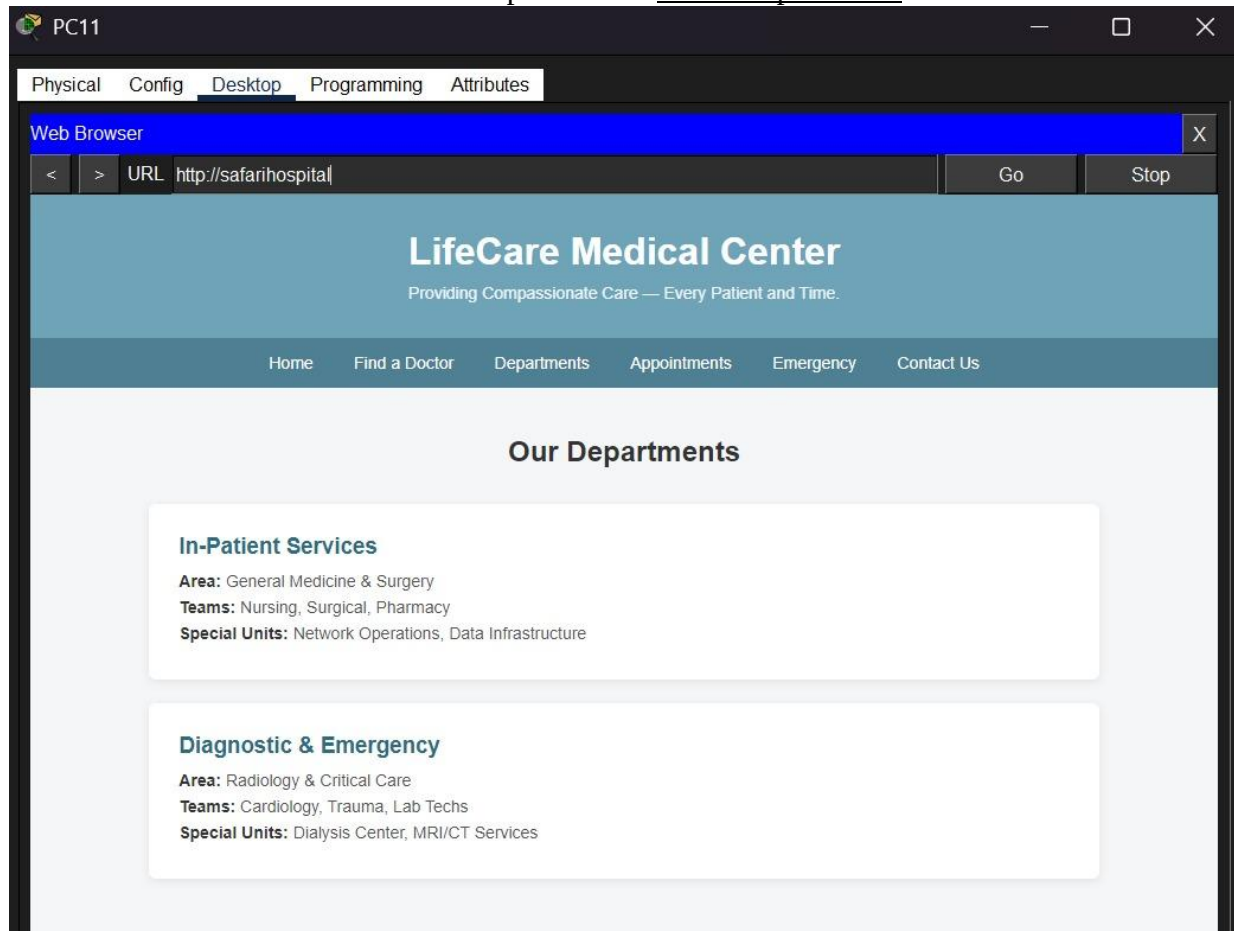
Save

Remove

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
Reception&Billing...	192.168.1....	192.168.1.98	192.168.1....	255.255.2...	6	0.0.0.0	0.0.0.0
GuestWifiArea2	192.168.1.81	192.168.1.98	192.168.1.87	255.255.2...	8	0.0.0.0	0.0.0.0
ClinicalArea1	192.168.1....	192.168.1.98	192.168.1....	255.255.2...	8	0.0.0.0	0.0.0.0
GeneralWardArea1	192.168.1.65	192.168.1.98	192.168.1.73	255.255.2...	6	0.0.0.0	0.0.0.0
ITDepVlan20	192.168.1.33	192.168.1.98	192.168.1.39	255.255.2...	24	0.0.0.0	0.0.0.0
ITDepVlan10	192.168.1.1	192.168.1.98	192.168.1.6	255.255.2...	25	0.0.0.0	0.0.0.0
PrivateWardVlan10	192.168.1....	192.168.1.98	192.168.1....	255.255.2...	25	0.0.0.0	0.0.0.0
PrivateWardVlan20	192.168.1....	192.168.1.98	192.168.1....	255.255.2...	24	0.0.0.0	0.0.0.0
serverPool	192.168.1....	192.168.1.97	192.168.1.0	255.255.2...	8	0.0.0.0	0.0.0.0

8.2 DNS & Web Server

The Web Server hosts the internal Hospital Portal www.hospital.local.



9. Network Security Implementation

9.1 Standard & Extended ACLs

ACLs are used to prevent Guest Wi-Fi users from accessing the IT Core Servers.

It Department:

```
%SYS-5-CONFIG_I: Configured from console by console

Router#show access-list
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#access-list 110 permit ip host 192.168.1.2 any
Router(config)#access-list 110 deny ip any host 192.168.1.2
Router(config)#access-list 110 permit ip any any
Router(config)#int fa0/0.10
Router(config-subif)#ip access-group 110 out
Router(config-subif)#
```

Copy

Paste

Private Ward:

```
Router(config)#
Router(config)#
Router(config)#
Router(config)#
Router(config)#access-list 120 permit ip host 192.168.1.130 any
Router(config)#access-list 120 deny ip any host 192.168.1.130
Router(config)#access-list 120 permit ip any any
Router(config)#int fa0/0.10
Router(config-subif)#ip access-group 120 out
Router(config-subif)#exit
Router(config)#
%SYS-5-CONFIG_I: Configured from console by console
```

Copy

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Restricting Webserver Access:

```
Serial0/0/0 from LOADING to FULL, Loading Done

Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#access-list 130 deny ip host 192.168.1.70 host
192.168.1.106
Router(config)#access-list 130 permit ip any any
Router(config)#int fa0/0
Router(config-if)#ip access-group 130 in
Router(config-if)#exit
Router(config)#|
```

Port Security:

To prevent unauthorized devices from plugging into the hospital wall jacks.

```
IOS Command
#PORT_SECURITY-2-PSECURE_VIOLATION: Security violation occurred, caused by MAC address 00D0..

Switch(config-if)#int fa6/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport port-security
Switch(config-if)#switchport port-security maximum 1
Switch(config-if)#switchport port-security mac-address 0060.47EA.6696
Found duplicate mac-address 0060.47ea.6696.
Switch(config-if)#switchport port-security violation shutdown
Switch(config-if)#int fa1/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport port-security
Switch(config-if)#switchport port-security maximum 1
Switch(config-if)#switchport port-security mac-address 0001.42BD.6B97
Switch(config-if)#switchport port-security violation shutdown
Switch(config-if)#int fa2/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport port-security
Switch(config-if)#switchport port-security maximum 1
Switch(config-if)#switchport port-security mac-address 0001.9762.C2B6
Switch(config-if)#switchport port-security violation shutdown
Switch(config-if)#int fa3/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport port-security
Switch(config-if)#switchport port-security maximum 1
Switch(config-if)#switchport port-security mac-address 0001.9767.57DC
Total secure mac-addresses on interface FastEthernet3/1 has reached maximum limit.
Switch(config-if)#switchport port-security maximum 2
Switch(config-if)#switchport port-security mac-address 0001.9767.57DC
Switch(config-if)#switchport port-security violation shutdown
Switch(config-if)#|
```

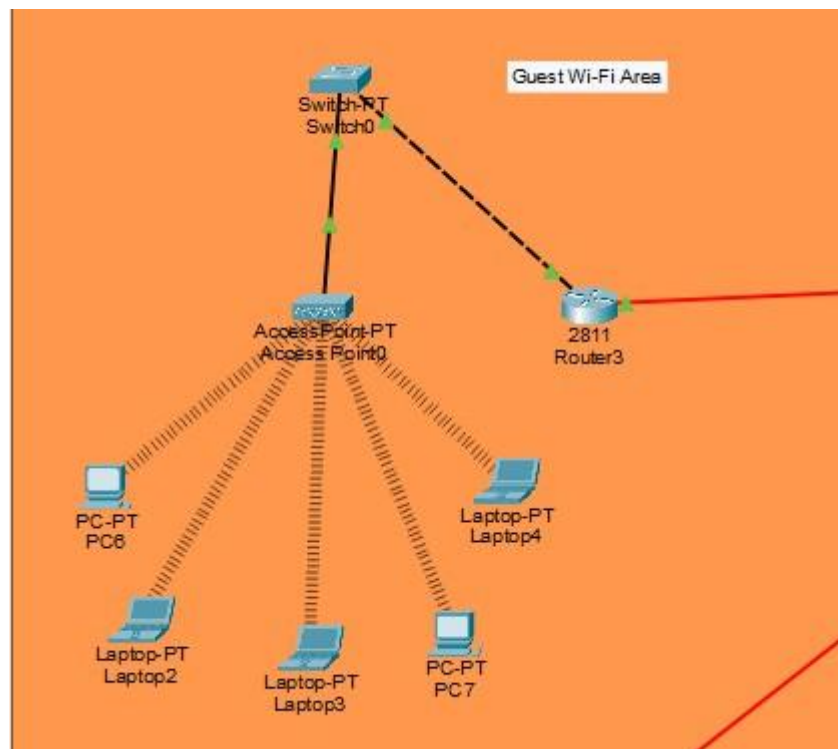
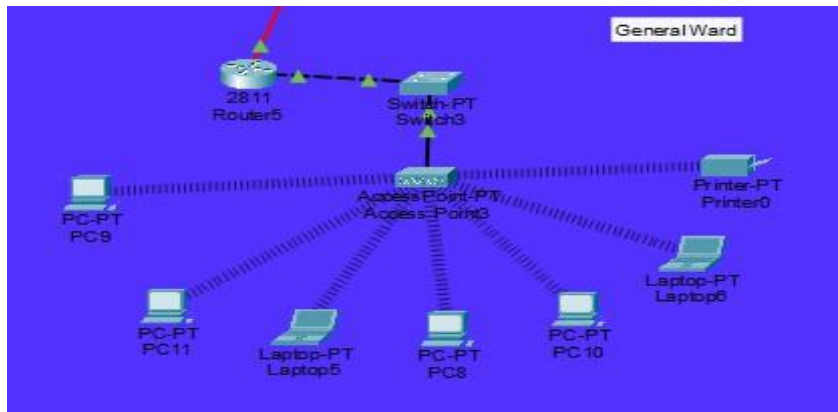
10. Wireless Network Design

10.1 AP Placement

Access Points are placed in the General Ward and Guest Area.

10.2 Security Protocol

WPA2-Enterprise/Personal was used to encrypt wireless traffic.



11. Performance & Quality of Service (QoS)

11.1 Latency Testing

We performed "Ping" tests between the furthest nodes (e.g., Guest PC to IT Manager).

11.2 Packet Loss Analysis

Result: 0% Packet loss during peak simulation.

```
Pinging 192.168.1.100 with 32 bytes of data:

Request timed out.
Reply from 192.168.187: bytes= 4 time=62
Reply from 192.168.187: bytes= 2 time=62
Reply from 192.168.187: bytes= 4 time=62
Reply from 192.168.187: bytes= 4 time=62

Ping statistics for 192.168.10.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 52ms, Maximum = 110ms, Average = 62ms
```


12. VPN Setting

Assigning IP address to interfaces:

```
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int se0/3/0
Router(config-if)#ip add 192.168.1.249 255.255.255.252
Router(config-if)#clock rate 56000
Router(config-if)#no shut

%LINK-5-CHANGED: Interface Serial0/3/0, changed state to down
Router(config-if)#
```

Creating access-list:

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/0, changed
state to up

Router(config-if)#exit
Router(config)#access-list 110 permit ip 192.168.1.0 0.0.0.127
192.168.1.128 0.0.0.127
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

Configuring ISAKMP:

```
Router(config)#
Router(config)#
Router(config)#crypto isakmp policy 10
Router(config-isakmp)#encryption aes
Router(config-isakmp)#hash sha256
^
% Invalid input detected at '^' marker.

Router(config-isakmp)#hash sha
Router(config-isakmp)#auth pre-share
Router(config-isakmp)#group 2
Router(config-isakmp)#lifetime 86400
Router(config-isakmp)#exit
Router(config)#crypto isakmp key VpnKey!23 address 192.168.1.250
Router(config)#crypto ipsec transform-set VPN-SET esp-aes esp-sha-hmac
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

Creating crypto map:

```
Router(config)#
Router(config)#
Router(config)#
Router(config)#crypto map VPN-MAP 10 ipsec-isakmp
% NOTE: This new crypto map will remain disabled until a peer
and a valid access list have been configured.
Router(config-crypto-map)#set peer 192.168.1.250
Router(config-crypto-map)#set transform-set VPN-SET
Router(config-crypto-map)#match address 110
Router(config-crypto-map)#
```

```
Router(config-crypto-map)#exit
Router(config)#
Router(config)#int se0/3/0
Router(config-if)#crypto map VPN-MAP
*Jan  3 07:16:26.785: %CRYPTO-6-ISA_KMP_ON_OFF: ISAKMP is ON
Router(config-if)#exit
Router(config)#
```

```
Router#ping 192.168.1.132

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.1.132, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 60/64/68 ms

Router#ping 192.168.1.166

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.1.166, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 3/17/22 ms

Router#
```

```
Router#ping 192.168.1.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.1.1, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 2/26/39 ms

Router#ping 192.168.1.67

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.1.67, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 38/49/61 ms

Router#
```

Conclusion:

In conclusion, the design and implementation of this Hospital Management System (HMS) network infrastructure successfully address the critical needs of a modern healthcare facility. By integrating a multi-branch architecture, the project demonstrates a high level of connectivity and data integrity.

Key takeaways from the project include:

Secure Connectivity: The implementation of a Site-to-Site IPSec VPN Tunnel ensures that sensitive patient records remain encrypted while traveling between the Head Office and the Branch Network.

Departmental Efficiency: Using VLANs to isolate departments like Finance, Nursing, and IT prevents unauthorized access and reduces broadcast traffic, enhancing overall network performance.

Scalability: The use of the OSPF dynamic routing protocol allows the hospital to easily expand its network to include more wards or satellite clinics without major reconfigurations.

Centralized Services: The Server-Side Network (SSS) provides a stable environment for essential services such as DHCP, DNS, and Web portals, which are vital for daily medical operations.

Proactive Security: Features such as Access Control Lists (ACLs) and Port Security provide multiple layers of defense against internal and external threats, ensuring HIPAA-level compliance.

This comprehensive network design provides a reliable foundation for digitizing healthcare services, improving patient care through technology, and maintaining the highest standards of data security.